

TAB 13



Board of Education Report

File #: Rep-134-23/24, **Version:** 1

Amendment to the Information Technology Services Strategic Execution Plan to Approve the Network Infrastructure Modernization Project

January 23, 2024

Information Technology Services

Action Proposed:

Approve amendment of the Information Technology Services (ITS) Strategic Execution Plan (SEP) to approve the definition, allocate funds, and take the associated action for the Network Infrastructure Modernization project, as described in Attachments A, B, and C.

The proposed actions are as follows:

- Allocate \$11,666,950 bond funding to complete the Network Infrastructure Modernization project.
- Approve execution of the Network Infrastructure Modernization project.

Background:

On August 24, 2021, the Board of Education approved the update to the SUP to integrate Measure RR funding and priorities (Brd. Rpt. 027-21/22), the next phase of the District's Bond Program, which will modernize, build, and repair school facilities to improve student health, safety, and educational quality. The Board's action approved the overarching goals and principles, funding sources, specific categories of need, and spending targets for the SUP. As proposed projects are developed, they are submitted to the Bond Oversight Committee (BOC) for consideration and the Board of Education for approval.

The SUP includes a spending target to “Upgrade and Equip [schools] with 21st Century Technology.” The intended outcome of the proposed project to be executed within this spending targets is as follows:

- Network Infrastructure Modernization project: This project will modernize the District’s network infrastructure, which connects all schools and offices to the District’s data center and Internet, by replacing equipment that will no longer be supported by the manufacturer with newer hardware that provides similar or greater functionality and capacity. \$11,666,950 Bond Funds (95%) and \$614,050 General Funds (5%). December 2026 anticipated completion.

If approved, ITS will provide quarterly updates on the project.

Bond Oversight Committee Recommendations:

The School Construction Bond Citizens’ Oversight Committee (BOC), at its January 11, 2024 meeting, considered this item. Staff has concluded that this proposed SEP Amendment will facilitate the implementation of the ITS SEP, and therefore it will not adversely affect the District’s ability to successfully complete the ITS SEP.

Expected Outcomes:

Staff anticipates that the Board will amend the ITS SEP to define the new project described herein and allocate

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\$11,666,950 of Bond Program funding. This project will enable the District to continue providing the stable IT Infrastructure required for teaching, learning and daily operations.

Furthermore, this project directly contributes to the achievement of the 2022-2026 Los Angeles Unified School District Strategic Plan goals of eliminating opportunity gaps (Pillar 1: Academic Excellence), creating welcoming learning environments (Pillar 2: Joy and Wellness), and modernizing infrastructure (Pillar 4: Operational Efficiency).

Board Options and Consequences:

Failure or delay of approval will postpone the execution and/or implementation of the proposed project and the associated benefits to the schools and students.

Policy Implications:

This action is consistent with the District's long-term goal to address unmet school needs as described in Proposition BB and Measures K, R, Y, Q, and RR.

Budget Impact:

The proposed project has a total budget of \$12,281,000, consisting of \$11,666,950 of Bond Program funding and \$614,050 of General Funds. The bond funds will be allocated from the "Upgrade and Equip with 21st Century Technology" SUP categories of need.

Upon completion of the project, \$200,000 in new funds will be needed to maintain the new equipment deployed. These funds will be planned for and budgeted out of the hardware/software and E-Rate budgets.

Student Impact:

The proposed project has the following outcomes and benefits that impact students:

- Support anticipated future growth of high-performance student and teacher computing devices.
- Deliver technology infrastructure that contributes to a safe and healthy learning environment.

Equity Impact:

Component	Score	Score Rationale
Recognition	4	Modernization of the District's network infrastructure provides improved and equal access to <u>all students</u> .
Resource Prioritization	4	Modernization of the District's network infrastructure provides improved and equal access to <u>all students</u> .
Results	4	This modernization is required to maintain the District's highly available network access in classrooms and on campuses, which directly contributes to eliminating opportunity gaps at District schools. Furthermore, this program contributes to other pillars and priorities of the 2022-2026 LAUSD Strategic Plan, including: Welcoming Learning Environments (Pillar 2: Joy and Wellness) and Modernize Infrastructure (Pillar 4: Operational Efficiency).
TOTAL	12	

Issues and Analysis:

As bandwidth to schools increases, the Network Infrastructure must also be increased. Based on current growth

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projections, the network infrastructure will need up to 4 times its current capacity to keep pace with learning needs. Furthermore, the existing network infrastructure equipment is reaching the end of its useful life. The equipment was purchased and installed nearly 10 years ago and will reach the end-of-support from manufacturers. At that point, there will be limited options for replacement of failed equipment and no security updates to patch vulnerabilities, leaving the District at risk of attack. The existing network equipment is operating at its maximum capacity. Additional capacity cannot be added to the existing network equipment to meet the District's growth needs.

Attachments:

Attachment A - Project Definition

Attachment B - Bond Oversight Committee Resolution

Attachment C - Presentation Slide Deck

Previously adopted Board report(s) referenced:

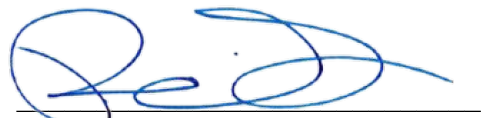
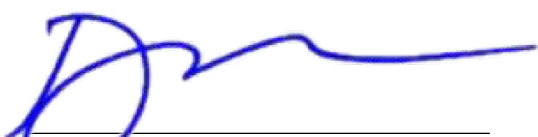
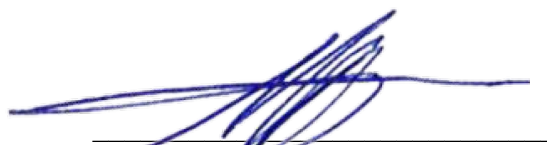
- In the background [Board Rpt. 027-21/22](https://drive.google.com/file/d/1zFCTkt0vcebc2R7IQM_bPi64HQMLSWBQ/view?)
<https://drive.google.com/file/d/1zFCTkt0vcebc2R7IQM_bPi64HQMLSWBQ/view?
dated: August 24, 2021.

Informatives:

None

Submitted:

12/1/23

RESPECTFULLY SUBMITTED,
ALBERTO CARVALHO
Superintendent**APPROVED BY:**
PEDRO SALCIDO
Deputy Superintendent
Office of the Deputy Superintendent**REVIEWED BY:**
DEVORA NAVERA REED
General Counsel☒ Approved as to form.**APPROVED & PRESENTED BY:**
SOHEIL KATAL
Chief Information Officer
Information Technology Services**REVIEWED BY:**
NOLBERTO DELGADILLO
Deputy Chief Business Officer, Finance
Office of the Chief Business Officer☒ Approved as to budget impact statement.

Program: School Network Infrastructure Modernization
Project: Network Infrastructure Modernization
Budget: \$12,281,000 total project budget: \$11,666,950 in Bond Funds and \$614,050 in General Funds

Introduction

As students increasingly rely on devices and online content for learning, the District's IT infrastructure must evolve accordingly. The network's availability, consistency, and high performance are crucial to our educational mission. To accommodate the growing number of student devices and the rising use of online applications both on campus and remotely, we must upgrade and refresh our Network Infrastructure.

The Network Infrastructure is the backbone of the District's network, linking all schools and offices to our data center and Cloud resources, and providing internet access for students, teachers, and staff. As schools require more bandwidth, there's a corresponding need to enhance the Network Infrastructure. With projected growth trends, our network infrastructure will need to expand fourfold to cater to evolving educational requirements.

Moreover, our existing network equipment is nearing its operational lifespan. Installed almost a decade ago, the first set of equipment is set to reach its end-of-support phase from manufacturers as early as June 2024. Once that happens, our options to replace malfunctioning equipment will be limited. It's also important to note that the current equipment is operating at full capacity, making it impossible to increase capacity to accommodate the District's expanding needs. Furthermore, without security updates to address vulnerabilities, the District becomes susceptible to cyber threats.

Scope

The project includes the following phases:

Phase	Scope
Phase 1: Node Modernization	Replace WAN core routers and optical equipment at 4 nodes.
Phase 2: Data Center Router & Firewall Modernization	- Replace data center routers and switches at 2 data centers. - Upgrade data center firewalls at 2 data centers.
Phase 3: Core Switches & WAN Router Modernization	- Upgrade core switches at 4 nodes. - Upgrade WAN routers that connect to WAN service provider.

Stabilization & Closeout	• Monitor & optimize performance. • Project closeout. • Update network documentation.
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Expected Outcomes

The Network Infrastructure Modernization Project will provide the following benefits and outcomes:

- Supports anticipated future growth of high-performance student and teacher digital learning needs.
- Delivers new network infrastructure equipment that will be supported by the manufacturer for the next 7-10 years.
- Improves space and power efficiency.
- Delivers a technology infrastructure that contributes to a safe and healthy learning environment.
- Installs firewalls with greater capacity to protect the internal District assets in the data centers.
- Ensures software updates for the network equipment will continue to be available to address any future security vulnerabilities identified in the products.
- Directly contributes to achievement of LAUSD 2022-2026 Strategic Plan goals to eliminate opportunity gaps (Pillar 1: Academic Excellence), create welcoming learning environments (Pillar 2: Joy and Wellness), and modernize infrastructure (Pillar 4: Operational Efficiency).

Schedule

The project will be completed by December 31, 2026.

SCHOOL CONSTRUCTION BOND CITIZENS' OVERSIGHT COMMITTEE**Margaret Fuentes, Chair**

LAUSD Student Parent

D. Michael Hamner, FAIA, Vice-Chair

American Institute of Architects

Dr. Samantha Rowles, Secretary

LAUSD Student Parent

Robert Campbell, Executive Committee

L.A. Co. Auditor-Controller's Office

Scott Pansky, Executive Committee

L.A. Area Chamber of Commerce

Neelura Bell

CA Charter School Association

Chad Boggio

L.A. Co. Federation of Labor AFL-CIO

Jeffrey Fischbach

CA Tax Reform Assn.

Aleigh Lewis

L.A. City Controller's Office

Patrick MacFarlane

Early Education Coalition

Jennifer McDowell

L.A. City Mayor's Office

Brian Mello

Assoc. General Contractors of CA

Dr. Clarence Montecarlo

Tenth District PTSA

William O. Ross IV31st District PTSA**Santa Ramirez (Alternate)**

Tenth District PTSA

Connie Yee (Alternate)

L.A. Co. Auditor-Controller's Office

Vacant

AARP

Joseph P. Buchman – Legal Counsel

Burke, Williams & Sorensen, LLP

Lori Raineri and Keith Weaver – Oversight**Consultants**Government Financial Services Joint
Powers Authority**Timothy Popejoy**

Bond Oversight Administrator

Perla Zitle

Bond Oversight Coordinator

RESOLUTION 2023-38**BOARD REPORT 134-23/24****RECOMMENDING BOARD APPROVAL FOR THE AMENDMENT OF THE INFORMATION TECHNOLOGY SERVICES STRATEGIC EXECUTION PLAN TO APPROVE THE NETWORK INFRASTRUCTURE MODERNIZATION PROJECT**

WHEREAS, District Staff proposes that the Board of Education amend the Information Technology Services (ITS) Strategic Execution Plan (SEP) to approve the definition, allocate funds, and take the associated actions for the Network Infrastructure Modernization project, as described in Attachments A, B, and C; and

WHEREAS, District Staff proposes the following actions:

- Allocate \$11,666,950 bond funding to complete the Network Infrastructure Modernization project.
- Approve execution of the Network Infrastructure Modernization project; and

WHEREAS, On August 24, 2021, the Board of Education approved the update to the SUP to integrate Measure RR funding and priorities (Brd. Rpt. 027-21/22), the next phase of the District's Bond Program, which will modernize, build, and repair school facilities to improve student health, safety and educational quality. The Board's action approved the overarching goals and principles, funding sources, specific categories of need, and spending targets for the SUP. As proposed projects are developed, they are submitted to the Bond Oversight Committee (BOC) for consideration and the Board of Education for approval; and

WHEREAS, The SUP includes a spending target to address “Upgrade and Equip [schools] with 21st Century Technology.” The intended outcome of the proposed project to be executed within this spending target is:

- Network Infrastructure Modernization project: This project will modernize the District’s network infrastructure, which connects all schools and offices to the District’s data center and Internet, by replacing equipment that will no longer be supported by the manufacturer with newer hardware that provides similar or greater functionality and capacity. \$11,666,950 Bond Funds (95%) and \$614,050 General Funds (5%)

RESOLUTION 2023-38**RECOMMENDING BOARD APPROVAL FOR THE AMENDMENT OF THE INFORMATION TECHNOLOGY SERVICES STRATEGIC EXECUTION PLAN TO APPROVE THE NETWORK INFRASTRUCTURE MODERNIZATION PROJECT**

If approved, ITS will provide quarterly updates on the Project; and

WHEREAS, District Staff anticipates that the Board will amend the ITS SEP to define the new project described herein and allocate \$11,666,950 of Bond Program funding. This project will enable the District to continue providing the IT Infrastructure required for teaching, learning and daily operations. Furthermore, this project directly contributes to the achievement of the 2022-2026 Los Angeles Unified School District Strategic Plan goals of eliminating opportunity gaps (Pillar 1: Academic Excellence), creating welcoming learning environments (Pillar 2: Joy and Wellness), and modernizing infrastructure (Pillar 4: Operational Efficiency); and

WHEREAS, the District Office of General Counsel has reviewed the proposed Project and determined that it may proceed to the Bond Oversight Committee for its consideration for recommendation to the Board of Education.

WHEREAS, District Staff has concluded this proposed ITS SEP amendment will facilitate Los Angeles Unified's ability to successfully complete the ITS SEP.

NOW, THEREFORE, BE IT RESOLVED THAT:

1. The School Construction Citizens' Bond Oversight Committee recommends that the Board of Education adopt the amendment of the Information Technology Services Strategic Execution Plan to approve the IT Infrastructure to Support Learning Program listed above as described in Board Report 134-23/24, a copy of which is attached hereto in the form it was presented to the BOC and is incorporated herein by reference.
2. This resolution shall be transmitted to the Los Angeles Unified School District Board of Education and posted on the Bond Oversight Committee's website.
3. The District is directed to track the above recommendation and to report on the adoption, rejection, or pending status of the recommendations as provided in section 6.2 of the Charter and Memorandum of Understanding between the Bond Oversight Committee and the District.

ADOPTED on January 11, 2024, by the following vote:

AYES: 14

ABSTENTIONS: 0

NAYS: 0

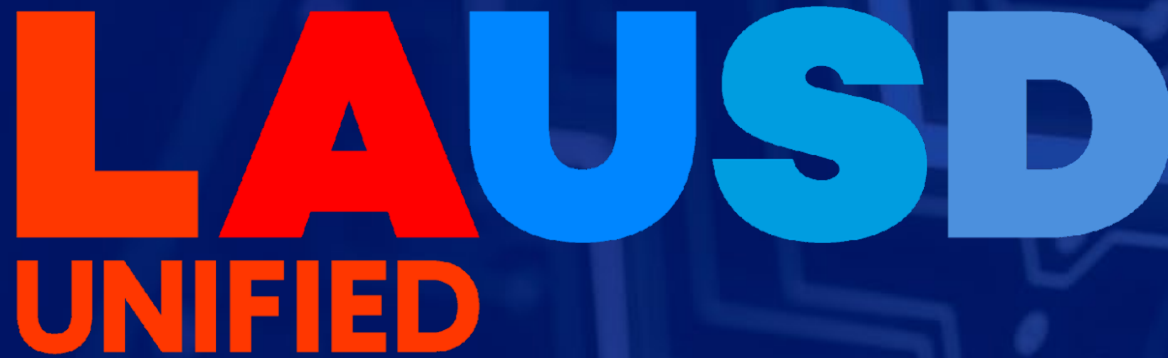
ABSENCES: 0

/Margaret Fuentes/

Margaret Fuentes
Chair

/Michael Hamner/

D. Michael Hamner
Vice-Chair



**Amendment to the ITS Strategic Execution
Plan to Approve the IT Infrastructure to
Support Learning Program**

Soheil Katal, Chief Information Officer
Dr. Frances Baez, Chief Academic Officer
January 23, 2024

Proposed Projects (slides 3 through 9)

As devices and online content become more and more essential to student learning and embedded in everyday life, the District's IT Infrastructure must be modernized to support this new learning paradigm

Proposed Projects (Slides 10 through 35)

1. Network Infrastructure Modernization (BOE #134-23/24) – Slides 11-18
2. School Network Systems Upgrade, Phase 2 (BOE # 118-23/24)– Slides 19-29
3. Intelligent School Network Controls (BOE #135-23/24)– Slides 30-35

Recommendation (Slide 36)

THE EXPERTS BELIEVE INSTRUCTIONAL TECHNOLOGY HAS THE CATALYTIC POWER TO TRANSFORM EDUCATION

U.S. Department of Education, Office of Education Technology:

Technology can be a powerful tool for transforming learning. It can help affirm and advance relationships between educators and students, reinvent our approaches to learning and collaboration, shrink long-standing equity and accessibility gaps, and adapt learning experiences to meet the needs of all learners

[Link: National Education Technology Plan Update](#)

&

Recommends building a robust technology infrastructure as a tool for equity

[Link: Building Robust Infrastructure as a Tool for Equity | by Office of Ed Tech | Medium](#)

Consortium for School Network:

Student access to robust digital tools is key to their success as 21st-century citizens.

Institute for Digital Transformation:

Identifies lack of IT infrastructure capable of enabling digital learning platforms as a top barrier to Digital Transformation

[Digital Strategy in Education \(institutefordigitaltransformation.org\)](https://www.institutefordigitaltransformation.org/)

STUDENTS USE LEARNING DEVICES TO ACCESS:



Learning Management System – Schoology



California Standardized Tests



Email and Collaboration Tools



E-curriculum



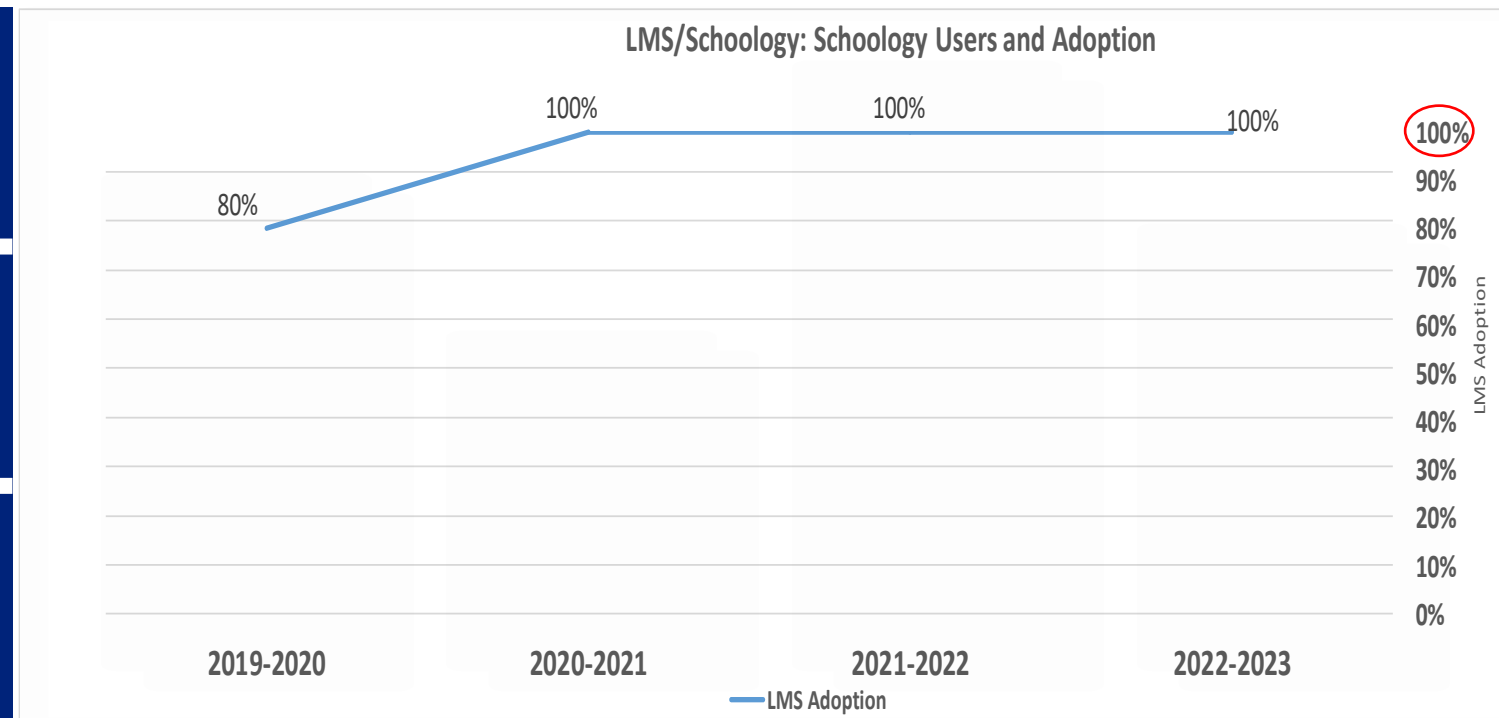
Credit recovery and online opportunities via Edgenuity



Video Conferencing

THE GROWTH IN ONLINE EDUCATION CONTENT: LEARNING MANAGEMENT SYSTEM (LMS) USAGE

- The LMS is a portal to student learning and engagement
- First introduced in 2016, the LMS was fully adopted by all District students by 2020-2021
- Teachers, Support staff, parents and others also utilize the LMS



INTERNET BANDWIDTH USAGE TRENDS & CAPACITY GROWTH FORECASTS

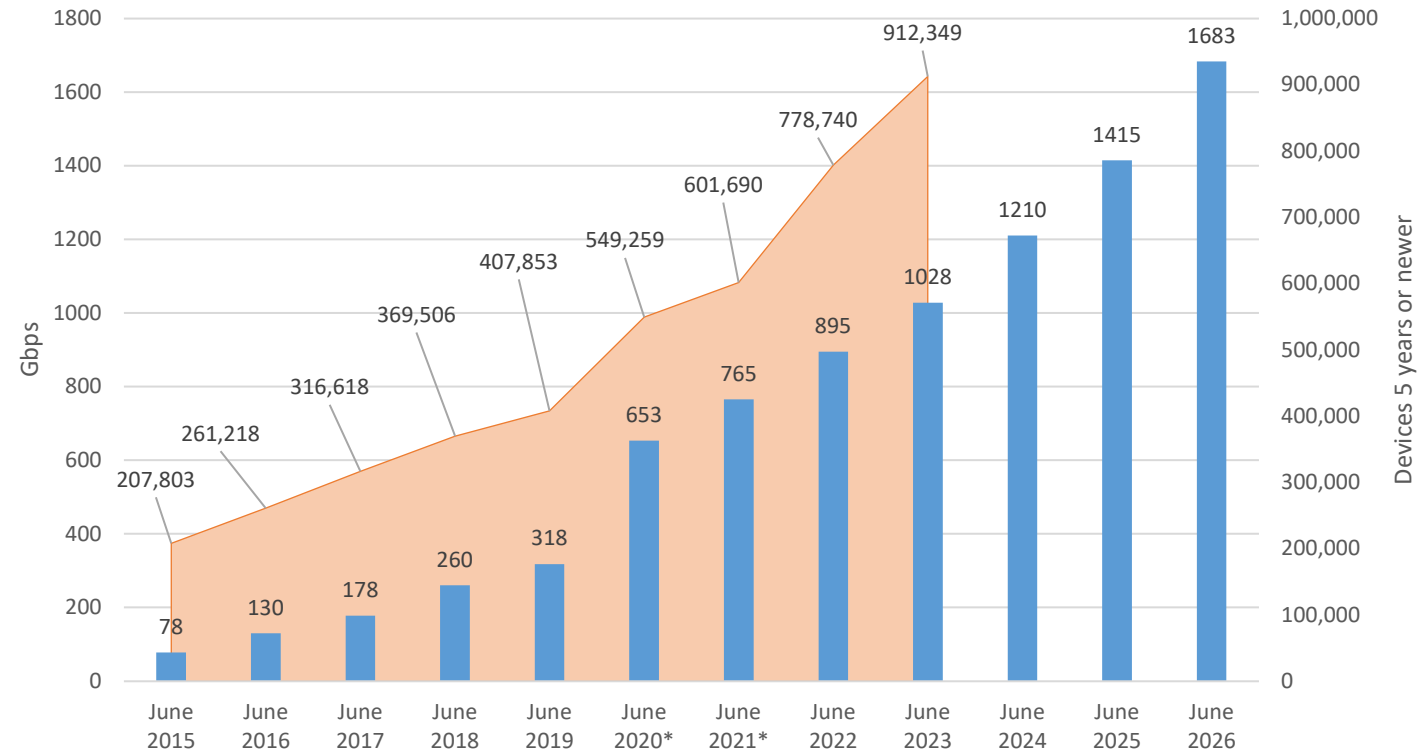
Best practice¹ guidelines for large K-12 districts

- 10 Gbps capacity per 1,000 students
- 1.4 Mbps per student

LAUSD bandwidth

- 4X increase in school computing devices
- 400 Gbps total for 435K students + staff
- 10 Gbps max capacity at each school
- 10X growth between 2015 and 2023
- 30% average annual growth rate
- Bandwidth usage is increasing due to online testing, increased digital content, and other instructional uses.

Site Bandwidth Usage & Estimated Growth



¹ State Educational Technology Directors Association & Federal Communications Commission recommendations

GROWTH IN SCHOOL-SITE LEARNING AND SUPPORT TECHNOLOGIES

- **Internet of Things (IoT)** describes the network of physical objects—“things”—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.
- **A school site's IoT-enabled universe can include:**
 - Instructional/ Teaching Devices, such as Virtual Reality Headsets, Interactive Whiteboards, Media Streamers, etc.
 - STEAM/STEM, Robotics, E-Sports program equipment
 - Building Automation Systems (BAS), such as Lighting Management, HVAC (Heating, Ventilating, and Air Conditioning), Low Voltage (Irrigation, Power Control, etc.), Photovoltaic (Solar, Battery, Energy Management, etc.), Food Service Kiosks, CCTV, Intrusion Alarms, Master Clocks, Bells, Speakers and Phones.



THE GROWTH IN SCHOOL-SITE LEARNING AND SUPPORT TECHNOLOGIES

Streaming
media adapter



eSports
computers & consoles



building
automation systems



Instructional Technology Growth / A New Learning Paradigm

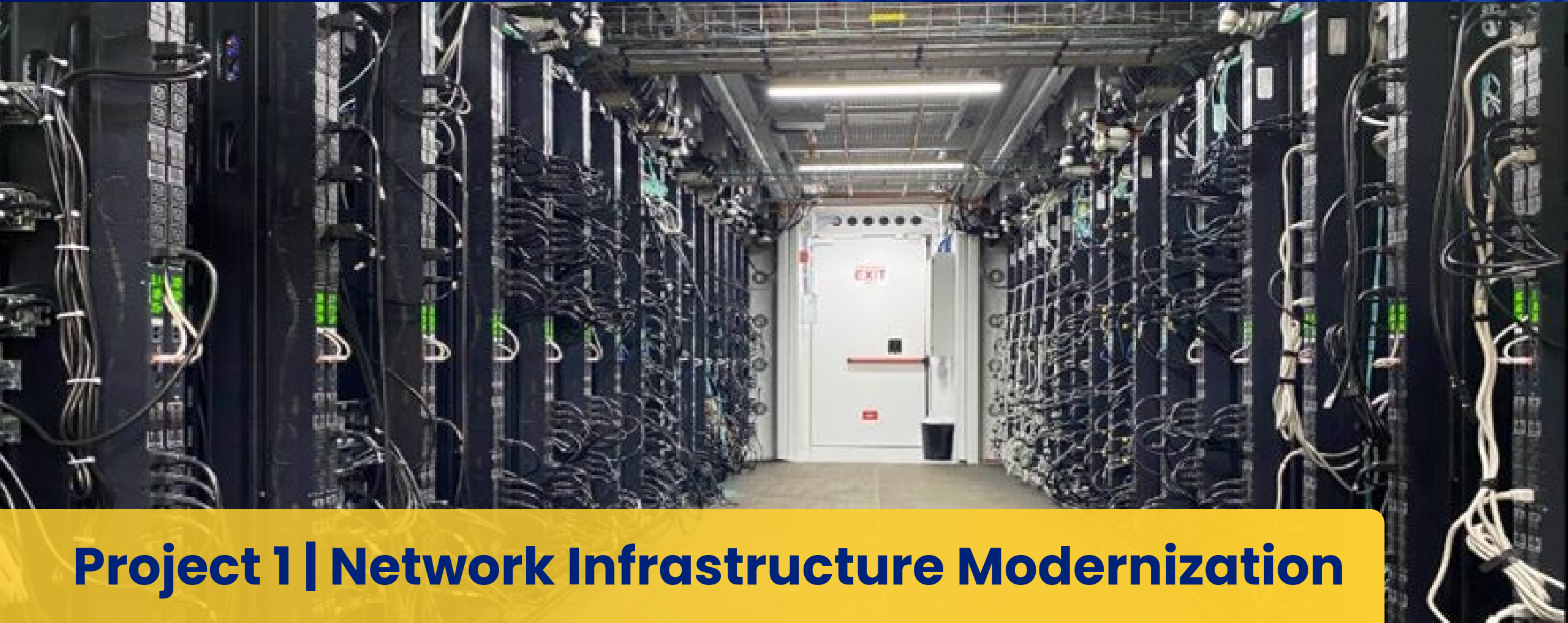
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The District's IT Infrastructure must be modernized to support the new learning paradigm.

PROPOSED PROJECTS

1) Network Infrastructure Modernization	Board Report 134-23/24	Slides 11-18
2) School Network Systems Upgrade, Ph 2	Board Report 118-23/24	Slides 19-29
3) Intelligent School Network Controls	Board Report 135-23/24	Slides 30-35



Project 1 | Network Infrastructure Modernization

Network Infrastructure Modernization: Outcomes & Benefits

278

Supports anticipated future growth of high-performance student and teacher computing devices

Delivers technology infrastructure that contributes to a safe and healthy learning environments

Delivers new network infrastructure equipment that will be supported by the manufacturer for the next 7-10 years

Improves space and power efficiency

Directly contributes to the achievement of the District's Goals to Eliminate Opportunity Gaps (Pillar 1: Academic Excellence), create Welcoming Learning Environments (Pillar 2: Joy and Wellness), and Modernize Infrastructure (Pillar 4: Operational Efficiency)

1

Academic Excellence

High-Quality Instruction

Enriching Experiences

Eliminating Opportunity Gaps

College and Career Readiness

2

Joy and Wellness

Welcoming Learning Environments

Whole-Child Well-Being

Strong Social-Emotional Skills

Outstanding Attendance

3

Engagement and Collaboration

Strong Relationships

Accessible Information

Leading for Impact

Honoring Perspectives

4

Operational Effectiveness

Data-Driven Decision-Making

Modernizing Infrastructure

Sustainable Budgeting

District of Choice

5

Investing in Staff

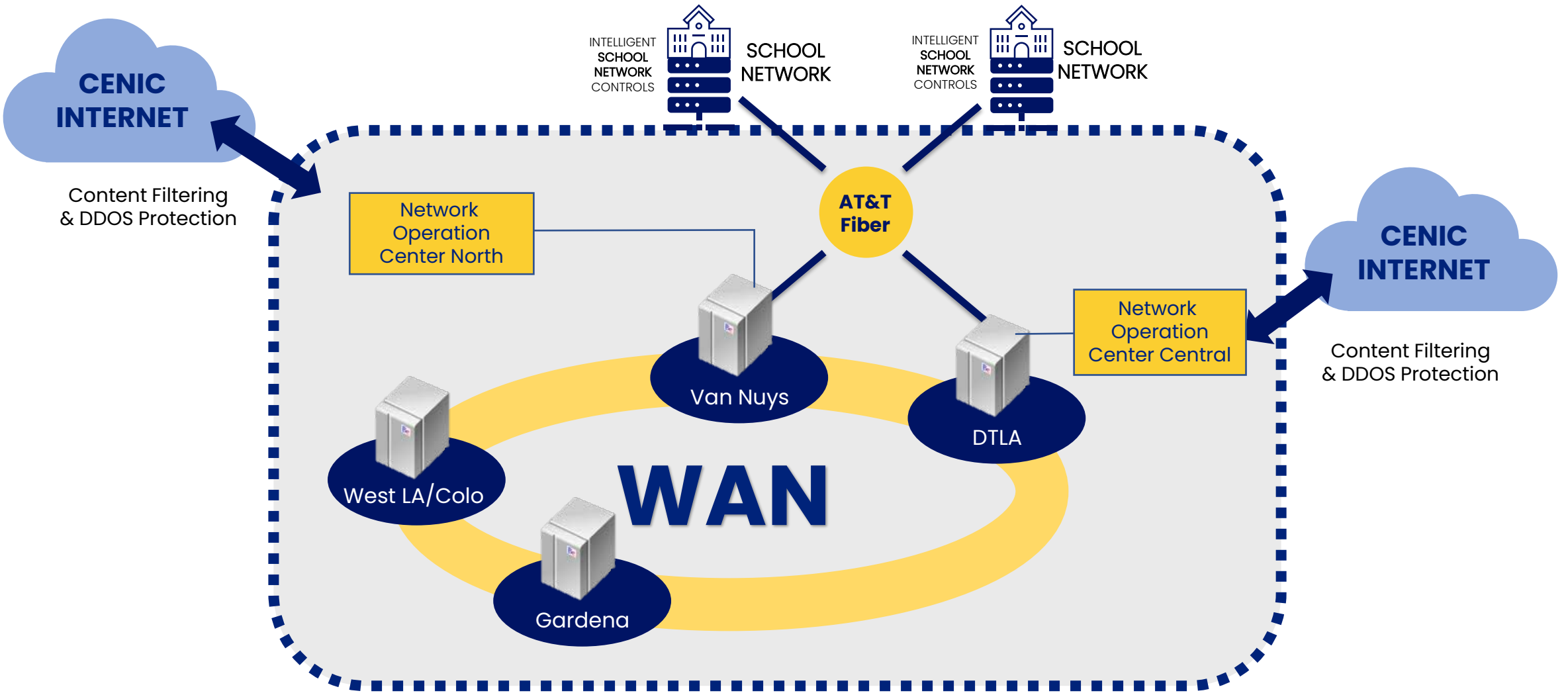
Diverse Workforce

Professional Learning

Staff Wellness

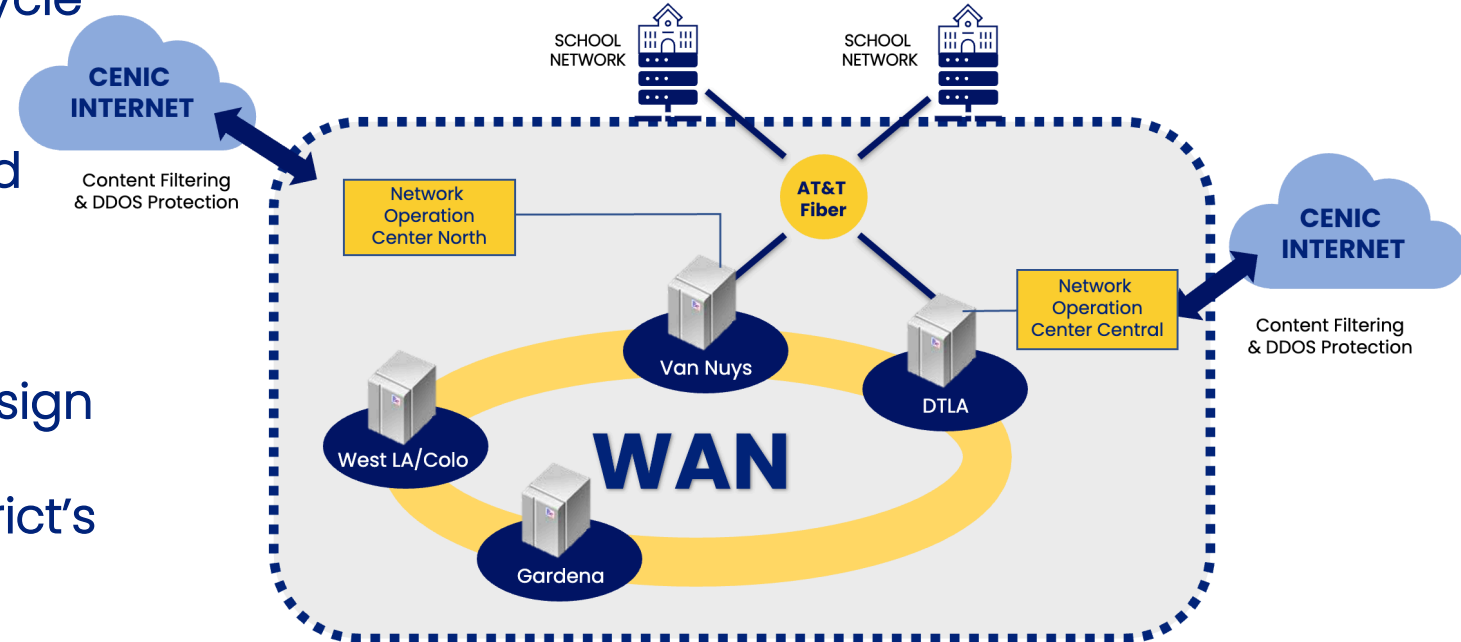
High Performance Standards

Los Angeles Unified's IT Infrastructure, High-Level Architecture



EXISTING CORE NETWORK HARDWARE IS:

- Nearing end of planned 7-10 year lifecycle
- Will reach End-of-Support from manufacturer starting in June 2024
 - Limited replacement parts for failed equipment
 - No security updates to patch vulnerabilities
- Operating at maximum equipment design capacity
- Cannot add capacity to meet the District's growth needs

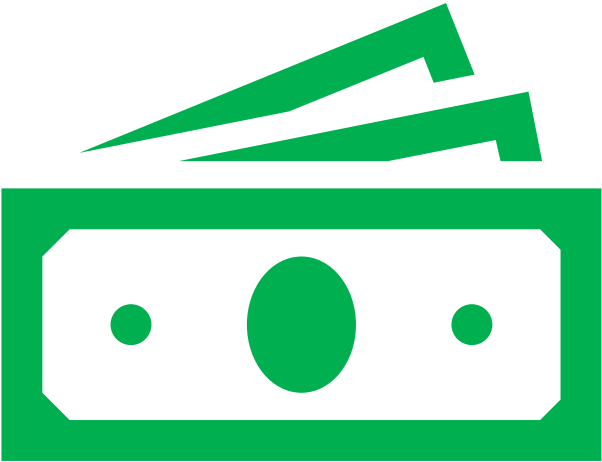


Network Infrastructure Modernization: Scope & Approach

Phase 1 Node Modernization	Phase 2 Data Center Routers & Firewall Modernization	Phase 3 Core Switches & WAN Routers Modernization	Stabilization & Closeout
Replace wide area network (WAN) core routers and optical equipment at 4 nodes Increases capacity from 400Gbps to 1.6Tbps	Replace data center routers and switches at 2 data centers Upgrade data center firewalls at 2 data centers	Upgrade core switches at 4 nodes Upgrade WAN routers that connect to WAN service provider	Monitor & optimize performance Project closeout Update network documentation



	2023				2024				2025				2026			
	Jan-Mar	Apr-Jun	Jul-Sep	Oct-Dec	Jan-Mar	Apr-Jun	Jul-Sep	Oct-Dec	Jan-Mar	Apr-Jun	Jul-Sep	Oct-Dec	Jan-Mar	Apr-Jun	Jul-Sep	Oct-Dec
Network Infrastructure Modernization																
Phase 1: Node Modernization																
Identify replacement products																
Procurement & Ordering																
Implementation & Go-Live																
Phase 2: Data Center Routers & Firewall Modernization																
Identify replacement products																
Procurement & Ordering																
Implementation & Go-Live																
Phase 3: Core Switch & WAN Router Modernization																
Identify replacement products																
Procurement & Ordering																
Implementation & Go-Live																
Stabilization & Closeout																



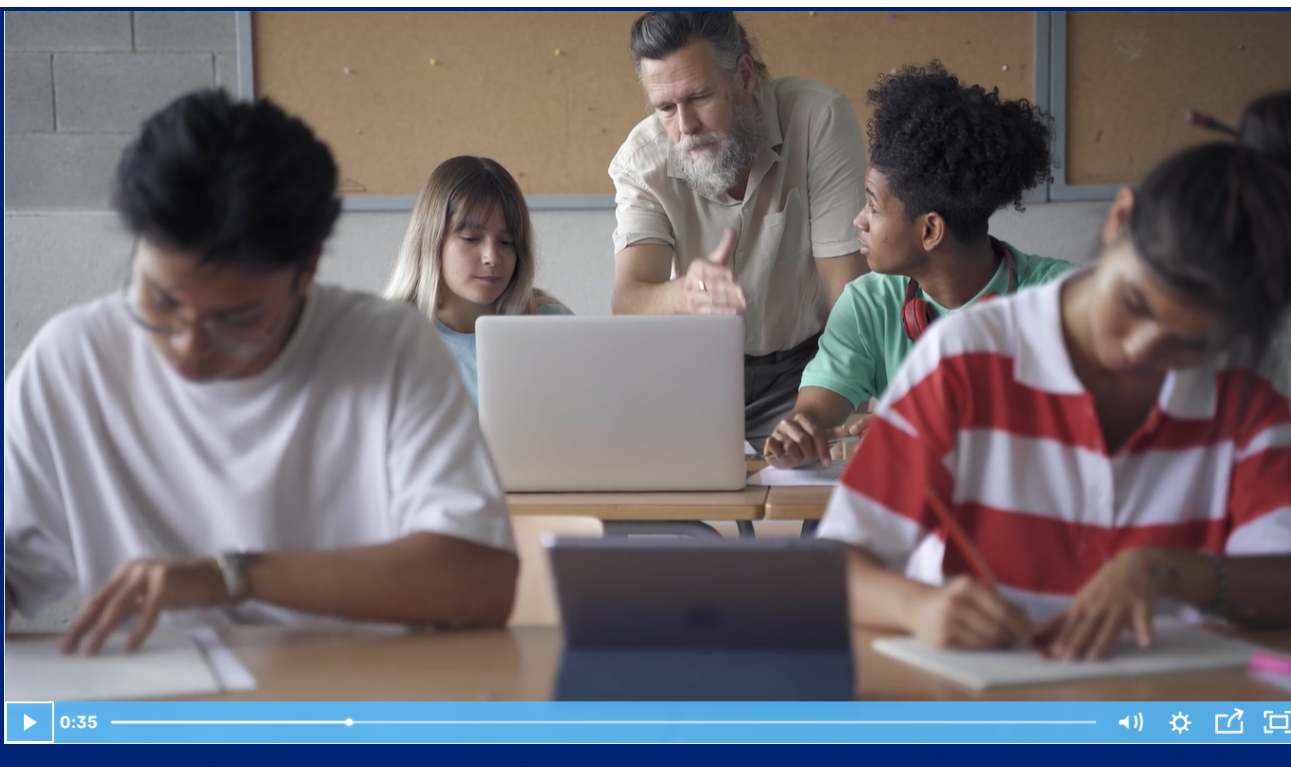
Cost Component	\$12,281,000*
Hardware **	\$11,569,000
Node Modernization (\$3,600,000)	
WAN & Data Center Routers (\$5,800,000)	
Core Switches Modernization (\$2,169,000)	
Labor & Professional Services	\$712,000

* Project will leverage 95% Bond Funds (\$11,666,950) and 5% General Funds (\$614,050)
** Funds may shift between phases depending upon several variables



Project 2 | School Network Systems Upgrade: Phase 2

School Network Systems Upgrades - School Feedback



“Due to the network upgrades classrooms are able to stay connected all day long.”

–Mathew Needleman (Principal – Melrose ES)

“It’s more accessible to today’s technology.”

–Frances Valadez (Principal – West Vernon ES)

“New systems reduce a lot of the [service] calls.”

–Paul Fallat II (Complex Project Manager)

“Teachers feel more connected to the office in an emergency.”

–Yadira Andrade (School Administrative Assistant – Bell HS)

“[New] phones are much more versatile than previous phones.”

–Andrea Kittelson (Principal – Walgrove ES)

Video: <https://lausd.wistia.com/medias/fh26c8ovtc>

School Network Systems Upgrade 286

Phase 2: Outcomes & Benefits

Improves speeds for streaming instructional content, utilizing applications such as Zoom, and accessing / downloading i.e., digital textbook content from the District's LMS.

Improves access to online content

Develops opportunities for career exploration, work-based learning, post-secondary pathways, and employment for students.

Promotes collaboration with DACE and ITS to provide apprenticeship opportunities for high school students to obtain a technical certification to qualify for employment as an Associate IT Electronics Communication technician.

Directly contributes to the achievement of the District's Goals of provide High-Quality Instruction, Eliminate Opportunity Gaps, and developing College and Career Readiness (Pillar 1: Academic Excellence)



School Network Systems Upgrade 287

Phase 2: Outcomes & Benefits

Improves communications throughout the schools and classrooms.

Provides a safer connection for increased security to enhance online and remote learning instruction.

Allows teachers, support staff, and administrators to connect with parents and other school stakeholders more easily and reliably.

Delivers new, reliable IT Infrastructure

Directly contributes to the achievement of the District's Goals to create Welcoming Learning Environments (Pillar 2: Joy and Wellness), build Strong Relationships (Pillar 3: Engagement & Collaboration), and Modernize Infrastructure (Pillar 4: Operational Efficiency)

1

Academic Excellence

High-Quality Instruction

Enriching Experiences

Eliminating Opportunity Gaps

College and Career Readiness

2

Joy and Wellness

Welcoming Learning Environments

Whole-Child Well-Being

Strong Social-Emotional Skills

Outstanding Attendance

3

Engagement and Collaboration

Strong Relationships

Accessible Information

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Honoring Perspectives

4

Operational Effectiveness

Data-Driven Decision-Making

Modernizing Infrastructure

Sustainable Budgeting

District of Choice

5

Investing in Staff

Diverse Workforce

Professional Learning

Staff Wellness

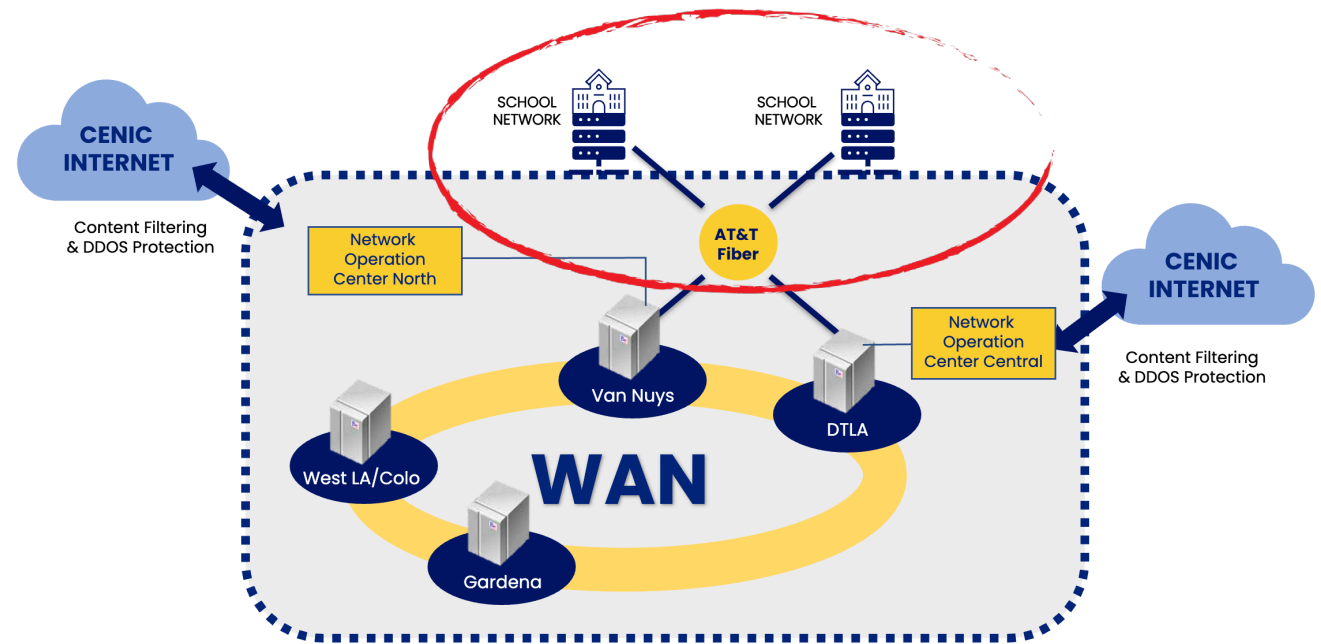
High Performance Standards

School Network Systems Upgrade, Phase 2: Challenges

288

Existing school network systems are:

- Beyond their planned lifecycles
 - Public address / intercommunications systems are 15–20+ years old
 - Telecommunications cabling infrastructure is 15–30+ years old
- Experiencing reliability & availability challenges



School Network Systems Upgrade, Phase 2: Roadmap

1

LAN Mod YR (2013–2015)

Measure K/R: 74 District school sites completed LAN Modernization.

2

Site Surveys (2015)

ITS conducted a two-year on-site IT infrastructure survey at all 713 schools.

3

S-A-F-E-T-I Criteria & Results (2017)

ITS Schools were assigned a score to assess their IT infrastructure conditions and were prioritized based on their level of repair need/urgency.

4

Identifying IT Upgrades (2017)

ITS developed Telecommunications Plan outlining IT upgrades needed at schools.

6

Upgrade Review (2020)

Development of a five-year plan To upgrade all schools to a Modern state.

5

Telecommunications Modernization Projects (TMP) (2018)

362 District school sites approved to receive combination of LAN, WLAN, PA, VoIP upgrades.

7

SNSU Ph 1 Project Approval (2021)

Group 1 (154 Sites) Projects In Progress.
Group 2 (108 Sites) :
• 21 Sites In Progress
• 87 Sites Awarded in 2023.

8

SNSU Ph 2 Project Approval (2024)

TCI Survey completed in 2023 to prioritize remaining group 3, 4, & 5 (349 Sites), which will go to Board for funding approval in January 2024

9

TMP 83/WLAN 160 Completion (2024)

Completion of Phone, PA, LAN, WLAN upgrades at 83 sites and Wireless Upgrades at 160 Sites (routers only) by June '24.

10

Next Refresh Cycle/SEP (2028)

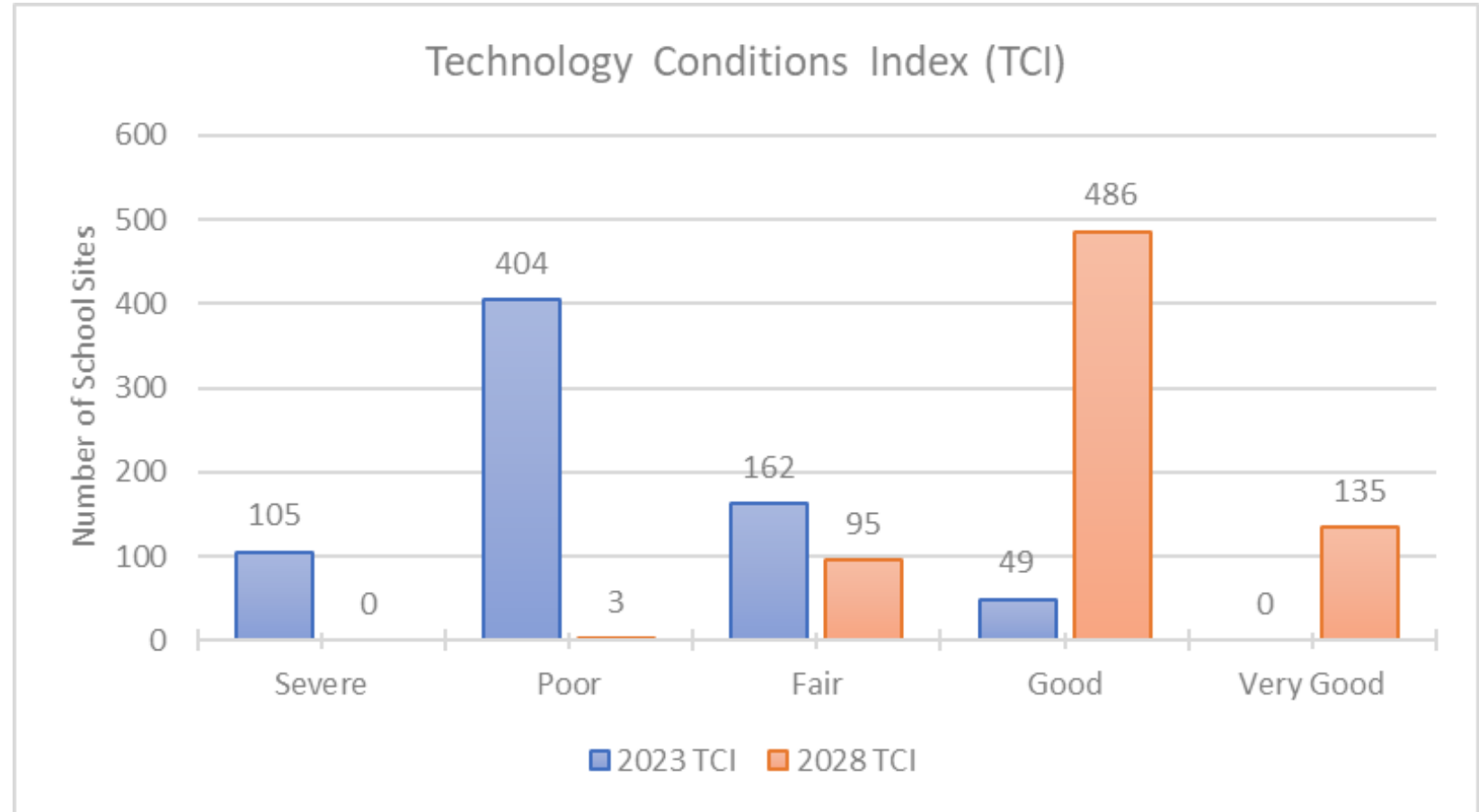
Approval of next Five-Year SEP to refresh LAN/WLAN for all schools including fiber optic cabling upgrades for 107 sites.

The Worst, First – Technology Conditions:

- 2023 Technology Condition Assessment data to prioritize the Phase 2 schools with the worst technology conditions based on age, system failures, and useful life.

The Distribution Ratio:

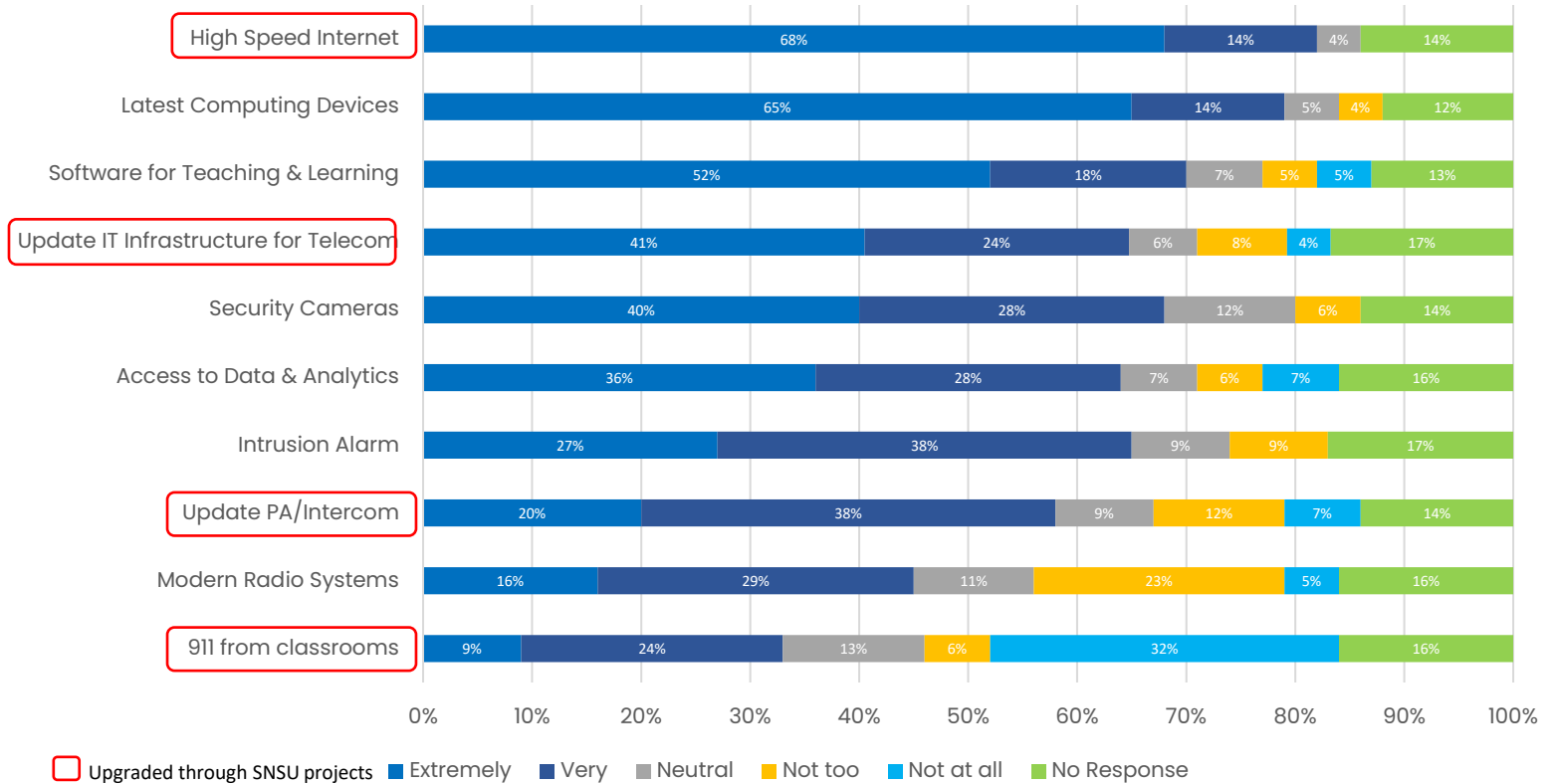
- Weighting additional factors such as the SENI score and ISTAR data along with technology conditions are applied to the Phase 2 schools list to achieve an equitable distribution for projects across the District.



School Leadership Feedback:

- Six (90-minute) Zoom sessions held with 54 participants to collect input through online polling tool and follow-up discussions.
- Participants included Region leadership and school administrators.
- Results prioritized “High Speed Internet” across all grade levels.

Importance of Technology Investments at Schools



School Network Systems Upgrades – Completions

LAN



NEW LAN



PA



NEW PA



PHONES



NEW PHONES



School Network Systems Upgrade, Phase 2 : Scope & Benefits

This project's scope includes surveying, planning, designing, procuring, installing, configuring, testing and acceptance of a new 10 Gigabit fiber infrastructure, LAN/WLAN, VoIP telephone system, and new integrated IP PA Intercommunications solution at the school sites.

Targeted Components

Project Output

Benefit to Students & Schools



Fiber Optic Cabling

10G fiber for improved capacity and bandwidth speeds

Improves access to online content and provides a safer connection for increased security to enhance online and remote learning instruction.



Local Area Network (LAN) / Wireless Local Area Network (WLAN)

Faster internet speeds to the classroom and **modernized** backup batteries

Improves speeds for streaming instructional content, utilizing applications such as Zoom, and accessing / downloading i.e., digital textbook content from the District's LMS.



Voice-over-Internet Protocol (VoIP) Phone

Enhanced calling service and integration with PA system with less risk of phone outages

Allows teachers, support staff, and administrators to connect with parents and other school stakeholders more easily and reliably.



Public Address (PA)

More reliable and integrated PA and phone systems

Improves public address communications throughout a school and school classrooms.

School Network Systems Upgrade, Phase 2 : Project Schedule

Phase 2 projects:

- Pending Board approval, project will commence in Q1 2024.

Phase 2 Schedule	2024				2025				2026				2027				2028			
	Jan-Mar	Apr-Jun	Jul-Sep	Oct-Dec	Jan-Mar	Apr-Jun	Jul-Sep	Oct-Dec	Jan-Mar	Apr-Jun	Jul-Sep	Oct-Dec	Jan-Mar	Apr-Jun	Jul-Sep	Oct-Dec	Jan-Mar	Apr-Jun	Jul-Sep	Oct-Dec
School Network Systems Upgrade, Phase 2																				
Group 3 (110 Sites)																				
Assess / Design																				
Installation																				
Group 4 (106 Sites)																				
Assess / Design																				
Installation																				
Group 5 (133 Sites)																				
Assess / Design																				
Installation																				

School Network Systems Upgrade, Phase 2 : Project Budget

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Phase 2 projects:

Group 3 | 110 Sites

Group 4 | 106 Sites

Group 5 | 133 Sites

Phase/Year	Budget
Phase 2	\$382,487,934*
Group 3	\$97,184,427
Group 4	\$131,628,394
Group 5	\$153,675,113

* Project will leverage 100% Bond Funds



Project 3 | Intelligent School Network Controls

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Intelligent School Network Controls: Outcomes & Benefits

Frees up high-value instructional time by allowing local school staff to quickly register smart school devices on the local network without requiring lengthy ITS processing delays.

Increases the quantity and type of usable smart devices for instructional purposes on the network.

Improves network accessibility and visibility and safeguards smart school devices from data breaches.

Complies with requirements to protect student and business information through network segmentation.

Provides analysis and insight into the performance and security of smart devices to sustain continued uptime for instructional use.

Directly contributes to the achievement of the District's Goals to provide High-Quality Instruction & Eliminate Opportunity Gaps (Pillar 1: Academic Excellence), and Modernize Infrastructure (Pillar 4: Operational Efficiency)

1

Academic Excellence

High-Quality Instruction

Enriching Experiences

Eliminating Opportunity Gaps

College and Career Readiness

2

Joy and Wellness

Welcoming Learning Environments

Whole-Child Well-Being

Strong Social-Emotional Skills

Outstanding Attendance

3

Engagement and Collaboration

Strong Relationships

Accessible Information

Leading for Impact

Honoring Perspectives

4

Operational Effectiveness

Data-Driven Decision-Making

Modernizing Infrastructure

Sustainable Budgeting

District of Choice

5

Investing in Staff

Diverse Workforce

Professional Learning

Staff Wellness

High Performance Standards

WHY DO SCHOOLS NEED INTELLIGENT SCHOOL NETWORK CONTROLS:

Intelligent School Network Controls allow IoT-enabled devices to quickly and securely connect to a school-site network. Without these Controls educators, students and facility managers may experience:

1. Limited number and type of smart devices capable of being used in the classroom.
2. Increased security risks due to lack of device segmentation.
3. Rejection of or delays in authorization for devices to use the network for communications.
4. Inability to compete in sanctioned eSports STEAM/STEM, and Robotics programs.

Intelligent School Network Controls: Scope / Approach

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Configure

school networks to support multiple categories/types of IoT devices.

Deploy

a device profiling system to categorize IoT devices for secure connectivity.

Create

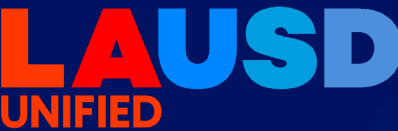
security policies to safeguard and accommodate different types of devices.

Deliver

new network services needed by modern classroom instruction.

Provide

a self-service portal to register devices that cannot be categorized.



Intelligent School Network Controls: Project Schedule

	2023				2024				2025			
	Jan-Mar	Apr-Jun	Jul-Sep	Oct-Dec	Jan-Mar	Apr-Jun	Jul-Sep	Oct-Dec	Jan-Mar	Apr-Jun	Jul-Sep	Oct-Dec
Intelligent School Network Controls												
Workstream 1 - Configure School Networks												
Workstream 2 - Develop Policies and Procedures												
Workstream 3 - Configure Device Registration Portal												
Workstream 4 - Deploy Device Profiling and Automation												
Stabilization & Closeout												

Intelligent School Network Controls: Project Budget

Intelligent School Network Controls	\$5,471,000*
Hardware	\$1,200,000
Software	\$1,842,000
Labor & Professional Services	\$2,417,000
Training	\$12,000

* Project will leverage 100% Bond Funds

IT Infrastructure to Support Learning: Recommendation

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ITS recommends amending the ITS Strategic Execution Plan to modernize the District's IT Infrastructure to Support Learning, utilizing a little over \$400M in existing bond & general funds, over the next 5 years.



Questions?